$$\boxed{p_0\, U(X, H_0) + (1-p_0)\, U(X, H=0) = p_1\, U(X-A, H_0) + (1-p_1)\, U(X-A, H=0)}$$

$$U(X-A, H_0) \approx U(X, H_0) - A\, U'(X, H_0)$$

$$U(X-A, H=0) \approx U(X, H=0) - A\, U'(X, H=0)$$

$f(x) = f(x_0 + \Delta) \approx f(x_0) + \Delta f'(x_0)$

$$p_0\, U(X, H_0) + (1-p_0)\, U(X, H=0) \approx U(X, H_0)\, p_1 + (1-p_1)\, U(X, H=0) - A\,\big(p\, U'(X, H_0) + (1-p)\, U'(X, H=0)\big)$$

$$\boxed{\frac{A}{\Delta p} = \frac{U(X, H_0) - U(X, H=0)}{p\, U'(X, H_0) + (1-p)\, U'(X, H=0)}}$$

	A	B
	c_A	c_B
different benefits	B_A	B_B

if benefits are equal → choose lowest cost ⇒ cost minimization

cost effectiveness

$\frac{c_A}{B_A}$ $\frac{c_B}{B_B}$ → cost per unit of benefit ⇒ choose the lowest ratio

Benefit B = effectiveness measure (blood pressure)

other benefits { blood pressure + better sleep } → α_1 Blood pressure + α_2 better sleep = index benefits of I

$\frac{c_A}{I_A}$ $\frac{c_B}{I_B}$ ← utility ⇒ cost utility analysis

$I = f(\text{blood pressure}, \text{sleep})$

$I = f(\underline{x})$

Quality Adjusted Life years: longevity - years of life
lifetime quality of life

$$QALY = \sum_{i=1}^{t} q_i \qquad \boxed{U = x_1^{\alpha_1} x_2^{\alpha_2}}$$

$\boxed{0{,}75 + 0{,}75 = 1 + 0{,}5}$

$\frac{c_A}{I_A \times \text{Mon value} \in}$ $\frac{c_B}{I_B \times \text{Mon value} \in}$ ← cost benefit analysis

VE

$I_A \times VE - C_A$

$I_B \times VE - C_B$

value of life → value of a year in perfect health

year 1x quality in year 1 + year 2x quality in year 2

ICER

incremental cost-effectiveness ratio

we start with option B

Question: shall we move to option A

$$ICER = \frac{C_A - C_B}{B_A - B_B} \leq \mu \text{ (threshold)}$$

43 000€

60 000€ per QALY

if it is OK to adopt (and pay)

80000€ > 60 000€ ⇒ do not adopt

$$C_A - C_B \leq \mu\,(B_A - B_B)$$

10 million

$$\boxed{\underbrace{\mu B_A - C_A}_{\text{cost benefit of option A}} \geq \underbrace{\mu B_B - C_B}_{\text{cost benefit of option B}}}$$

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measure of benefit

benefits? → may depend on the specific intervention

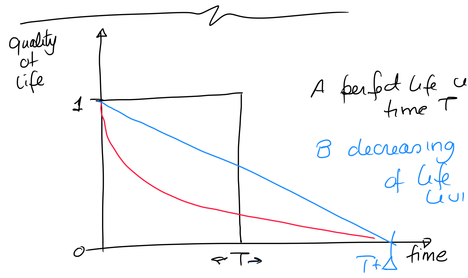
→ usually more than one

to have general and comparable across interventions

- time of life - longevity
- quality of life

} QALY

$I = f(x_1, \dots, x_n)$ n benefits



A perfect life until time T

B decreasing quality of life but living longer

• chronic condition t years 50 years

p — perfect health $p = 0{,}999$, $\boxed{0{,}95}$ 0,9

1-p — death

→ chronic 1 year (death afterwards)

p — perfect health 0,95, 0,8, 0,7 $\boxed{0{,}5}$

1-p — death

0,95 is a better health status than 0,5